

Linear Regression Using Top 6 Principal Components

1. Load Libraries

```
library(ggplot2)
library(dplyr)
library(car)      # for VIF
library(caret)    # for RMSE and data splitting
```

2. Load and Filter the Data

```
# Load cleaned Boston dataset
data <- read.csv("../data/boston_cleaned.csv")

# Check structure
str(data)
```

```
## 'data.frame':  506 obs. of  13 variables:
## $ CRIM      : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
## $ ZN        : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
## $ INDUS     : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
## $ CHAS      : int   0 0 0 0 0 0 0 0 0 0 ...
## $ NOX       : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
## $ RM        : num  6.58 6.42 7.18 7 7.15 ...
## $ AGE       : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
## $ DIS       : num  4.09 4.97 4.97 6.06 6.06 ...
## $ RAD       : int   1 2 2 3 3 3 5 5 5 5 ...
## $ TAX       : int  296 242 242 222 222 222 311 311 311 311 ...
## $ PTRATIO   : num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
## $ LSTAT     : num  4.98 9.14 4.03 2.94 5.33 ...
## $ MEDV      : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

3. Principal Component Analysis (PCA)

```
# Run PCA on standardized predictors
pr.out <- prcomp(data %>% select(-MEDV), scale. = TRUE)

# Extract top 6 principal components
pc_data <- as.data.frame(pr.out$x[, 1:6])
pc_data$MEDV <- data$MEDV
```

4. Split Data (80/20)

```
# Set seed for reproducibility
set.seed(2641)

# Split into training (80%) and testing (20%)
splitIndex <- createDataPartition(pc_data$MEDV, p = 0.8, list = FALSE)
train <- pc_data[splitIndex, ]
test <- pc_data[-splitIndex, ]
```

5. Fit Linear Regression Model

```
# Fit model on training data
PCReg <- lm(MEDV ~ ., data = train)
summary(PCReg)
```

```
##
## Call:
## lm(formula = MEDV ~ ., data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.646  -2.769  -0.680   1.785  33.106
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  22.4982     0.2665  84.416 < 2e-16 ***
## PC1         -2.2618     0.1109 -20.397 < 2e-16 ***
## PC2          2.7613     0.2194  12.584 < 2e-16 ***
## PC3          3.1639     0.2437  12.982 < 2e-16 ***
## PC4          1.7469     0.2787   6.268 9.47e-10 ***
## PC5         -1.4631     0.2982  -4.906 1.36e-06 ***
## PC6          0.3261     0.3606   0.905  0.366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.37 on 400 degrees of freedom
## Multiple R-squared:  0.6649, Adjusted R-squared:  0.6599
## F-statistic: 132.3 on 6 and 400 DF, p-value: < 2.2e-16
```

6. Predict & Evaluate on Test Set

```
# Predict on test set
predictions <- predict(PCReg, newdata = test)

# Calculate SSE and SST
SSE <- sum((test$MEDV - predictions)^2)
ybar <- mean(train$MEDV)
```

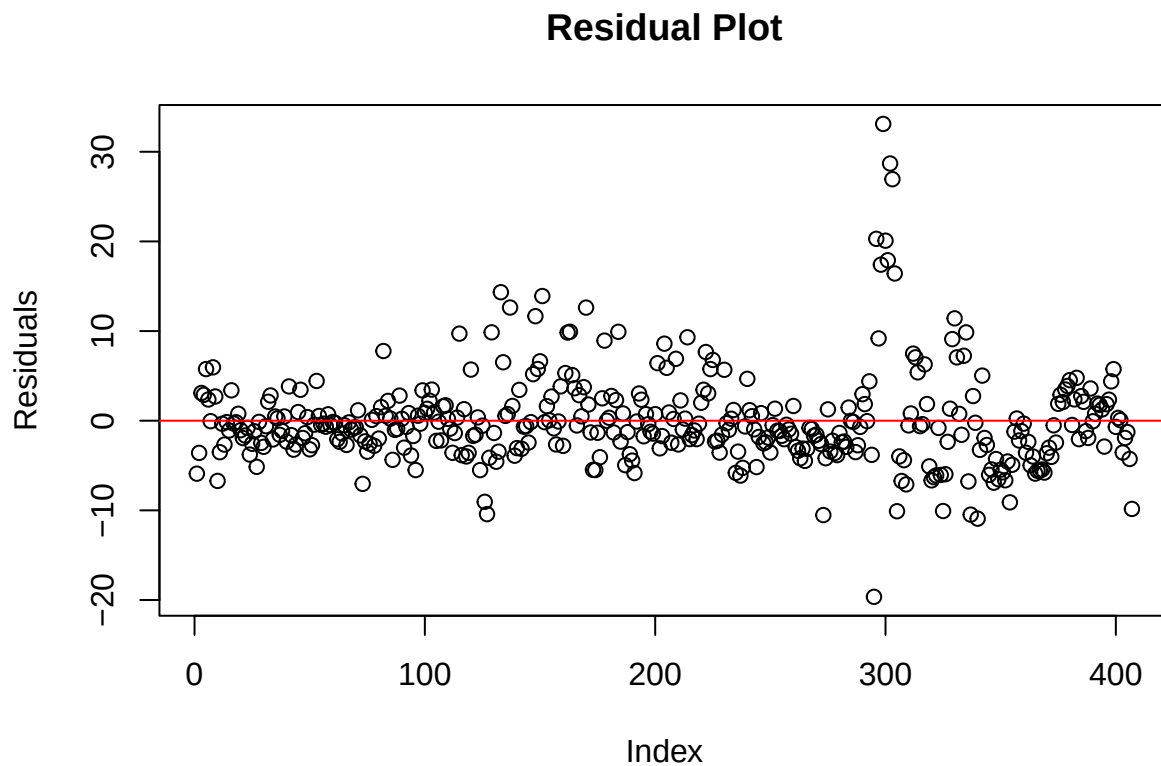
```
SST <- sum((test$MEDV - ybar)^2)

# Out-of-sample R-squared
R2_out <- 1 - SSE/SST
cat("Out-of-sample R2:", round(R2_out, 4))
```

```
## Out-of-sample R2: 0.7946
```

7. Residual Plot

```
# Residual plot
plot(PCReg$residuals, main = "Residual Plot", ylab = "Residuals", xlab = "Index")
abline(h = 0, col = "red")
```



8. Model Validation

a. T-tests and F-test (from `summary()`)

```
summary(PCReg) # includes t-values and F-statistic
```

```
##
## Call:
## lm(formula = MEDV ~ ., data = train)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -19.646  -2.769  -0.680   1.785  33.106
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## Coefficients:
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## Residual standard error: 5.37 on 400 degrees of freedom
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```

b. VIF (Variance Inflation Factor)

```
vif(PCReg) # check multicollinearity
```

```
##      PC1      PC2      PC3      PC4      PC5      PC6
## 1.001776 1.001932 1.002911 1.001193 1.001686 1.000678
```

Summary

- The model uses **PC1–PC6** which together explain ~93–94% of variance in the predictors.
- The **out-of-sample R^2** is 0.7946, showing how well the model generalizes.
- **No multicollinearity** is expected since PCs are orthogonal (VIF ~ 1).
- Model significance assessed via **t-tests** and **F-test**.